

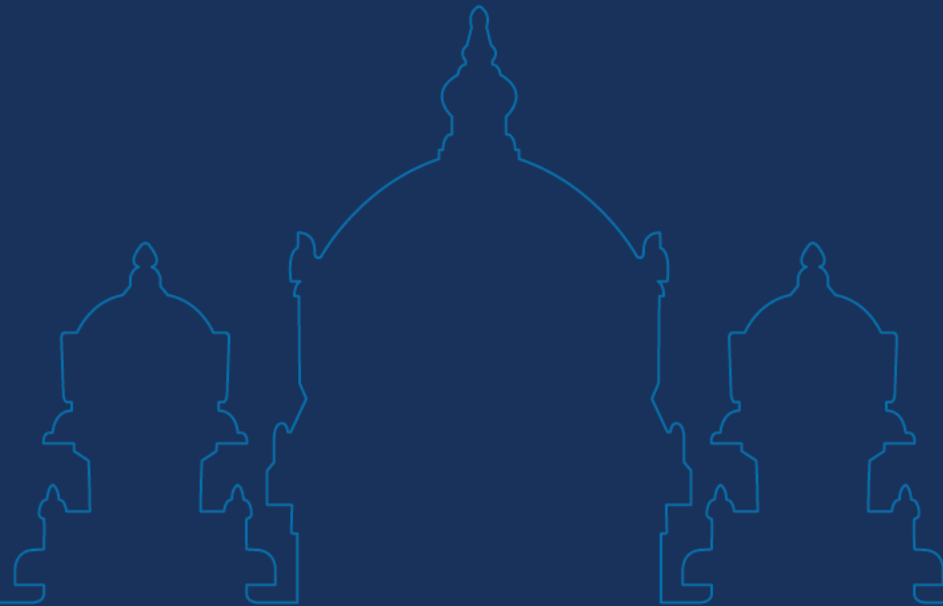
Data Science and Visualization

Unit II DESCRIPTIVE STATISTICS



KCED

Learning Augmented



Unit II – Descriptive Statistics

Advanced Attribute Types - Measures of Central Tendency - Measures of Dispersion - Skewness and Kurtosis - Visual Outlier Identification - Missing Data Visualization - Exploratory Data Analysis (EDA) Techniques - Histograms, KDEs, Box Plots, and Violin Plots.

CO2

Perform Exploratory Data Analysis by applying descriptive statistics and univariate/bivariate visualizations to uncover data distributions, patterns, and relationships, while analyzing data quality through the detection of outliers and exploration of missing value patterns.

Unit II – Descriptive Statistics

Deep Dive into Data/Attribute Types: Nominal-Ordinal-Binary-Numeric -Discrete vs. Continuous-Asymmetric Attributes-Measurement Scales - Basic Statistical Descriptions: Measures of Central Tendency -Mean-Median-Mode - Measures of Dispersion -Range-Quartiles-Variance-Standard Deviation-IQR - Understanding Skewness and Kurtosis -Conceptual - Data Cleaning Context: Visual identification of outliers - Visualizing Missing Data –Types of Missingness- Exploratory Data Analysis Visualization Techniques-Univariate: Histograms with Kernel Density Estimation-Box Plots -Violin Plots.

CO2

Perform Exploratory Data Analysis by applying descriptive statistics and univariate/bivariate visualizations to uncover data distributions, patterns, and relationships, while analyzing data quality through the detection of outliers and exploration of missing value patterns.

Descriptive Statistics

Measure of Central Tendency

- 1) Mean
- 2) Median
- 3) Mode

Measure of Variability / Dispersion

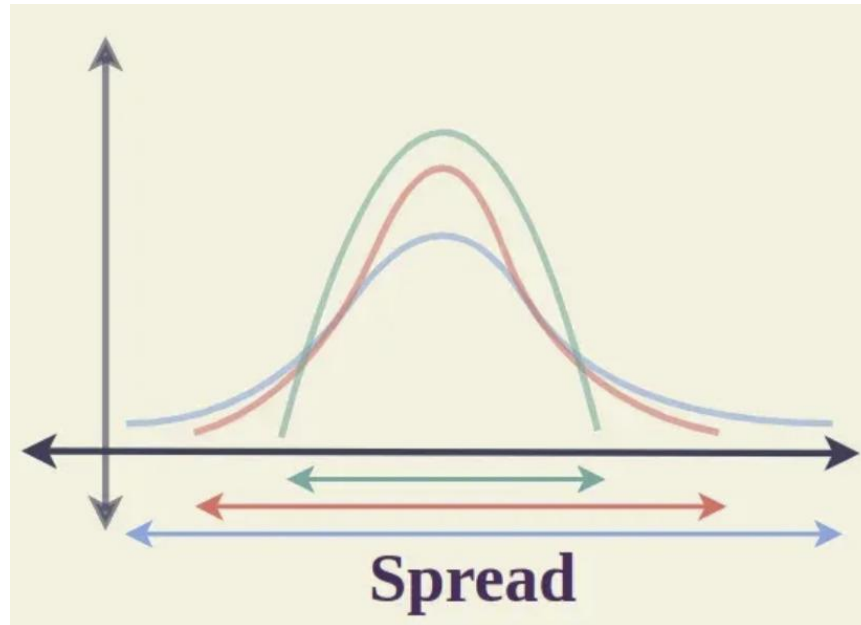
- 1) Range
- 2) IQR
- 3) Variance
- 4) Std. Dev.

Shape / Freq. Distribution

- 1) Skewness
- 2) Kurtosis

Measures of Dispersion

- Dispersion refers to the **degree to which data values in a dataset are spread out or scattered** around an average (such as the mean or median).
- It helps us understand the **variability or consistency within the data** — whether the values are **closely grouped or widely spread**.



Different spreads show varying levels of data dispersion

Dispersion ?



Spread / Scatter of Data

- Shows the extent to which a distribution of given data is stretched or squeezed.

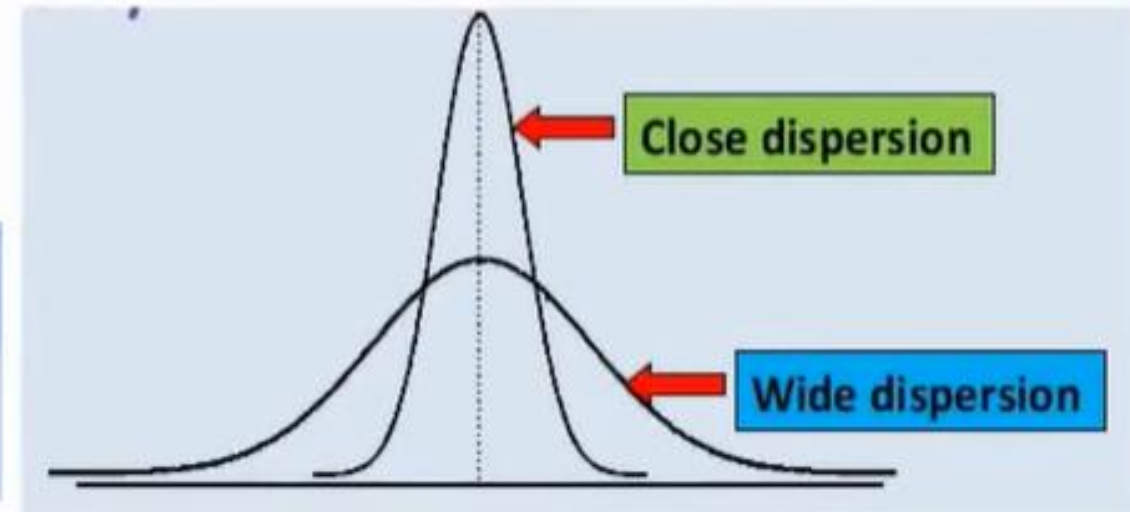
What does it tells us ?

Variability of data.

(How much homogeneous / heterogeneous the data is.)

Or, How squeezed or scattered a variable is.

Or, How much a data is spread around a central value (mean).



Measures of Dispersion

The most common measures of dispersion include:

- **Range** – the **difference** between the maximum and minimum values.
- **Variance** – the **average of the squared deviations** from the mean.
- **Standard Deviation** – the **square root of the variance**, showing dispersion in the same units as the data.

1.) Range

$$\text{RANGE} = \text{MAX} - \text{MIN}$$

- Simplest Measure

4, 5, 1, 3, 7 $\xrightarrow{\text{Range}}$ $7 - 1 = 6$

- Sensitive to Outliers 

4, 5, 1, 3, 7, 98 $\xrightarrow{\text{Range}}$ $98 - 1 = 97$

An Outlier

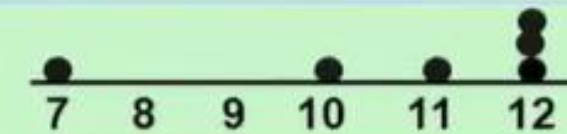
Higher Range

- Can be used for Small-Sized data.

- Ignores the way data is distributed 

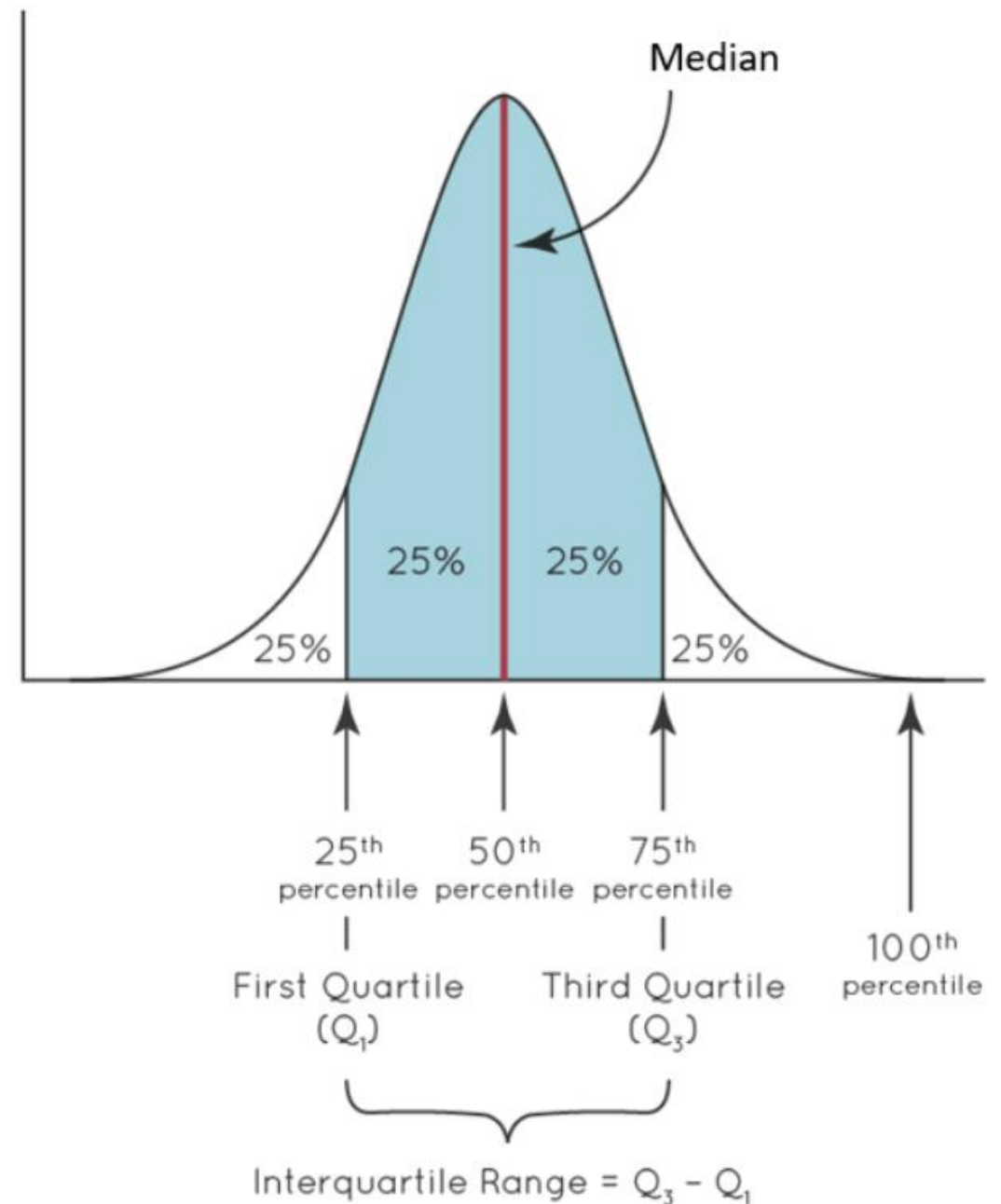


$$\text{Range} = 12 - 7 = 5$$



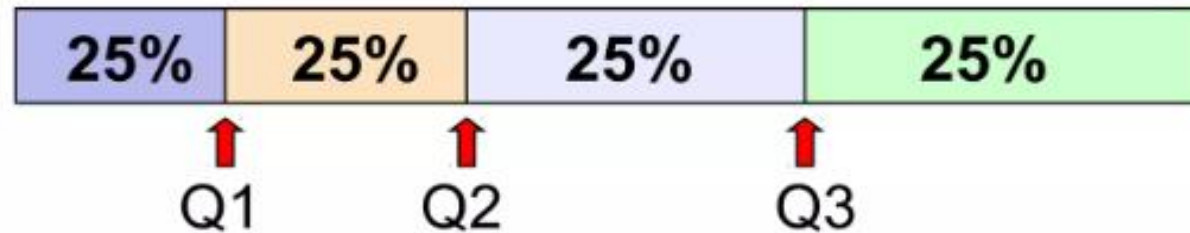
$$\text{Range} = 12 - 7 = 5$$

Quartiles & Percentiles



Quartiles

- Quartiles split the ranked data into 4 segments with an equal number of values per segment



- The first quartile, Q_1 , is the value for which 25% of the observations are smaller and 75% are larger
- Q_2 is the same as the median (50% are smaller, 50% are larger)
- Only 25% of the observations are greater than the third quartile

Quartiles

- Example: Find the first quartile

Sample Data in Ordered Array: 11 12 13 16 16 17 18 21 22

(n = 9)

Q_1 is in the $9/4 = 2.25$ position of the ranked data

so $Q_1 = 12.25$

Q_1 and Q_3 are measures of noncentral location
 Q_2 = median, a measure of central tendency

Quartiles

Find a quartile by determining the value in the appropriate position in the ranked data, where

First quartile position: $Q_1 = n/4$

Second quartile position: $Q_2 = n/2$ (the median position)

Third quartile position: $Q_3 = 3n/4$

where n is the number of observed values

2.) IQR (Interquartile Range)

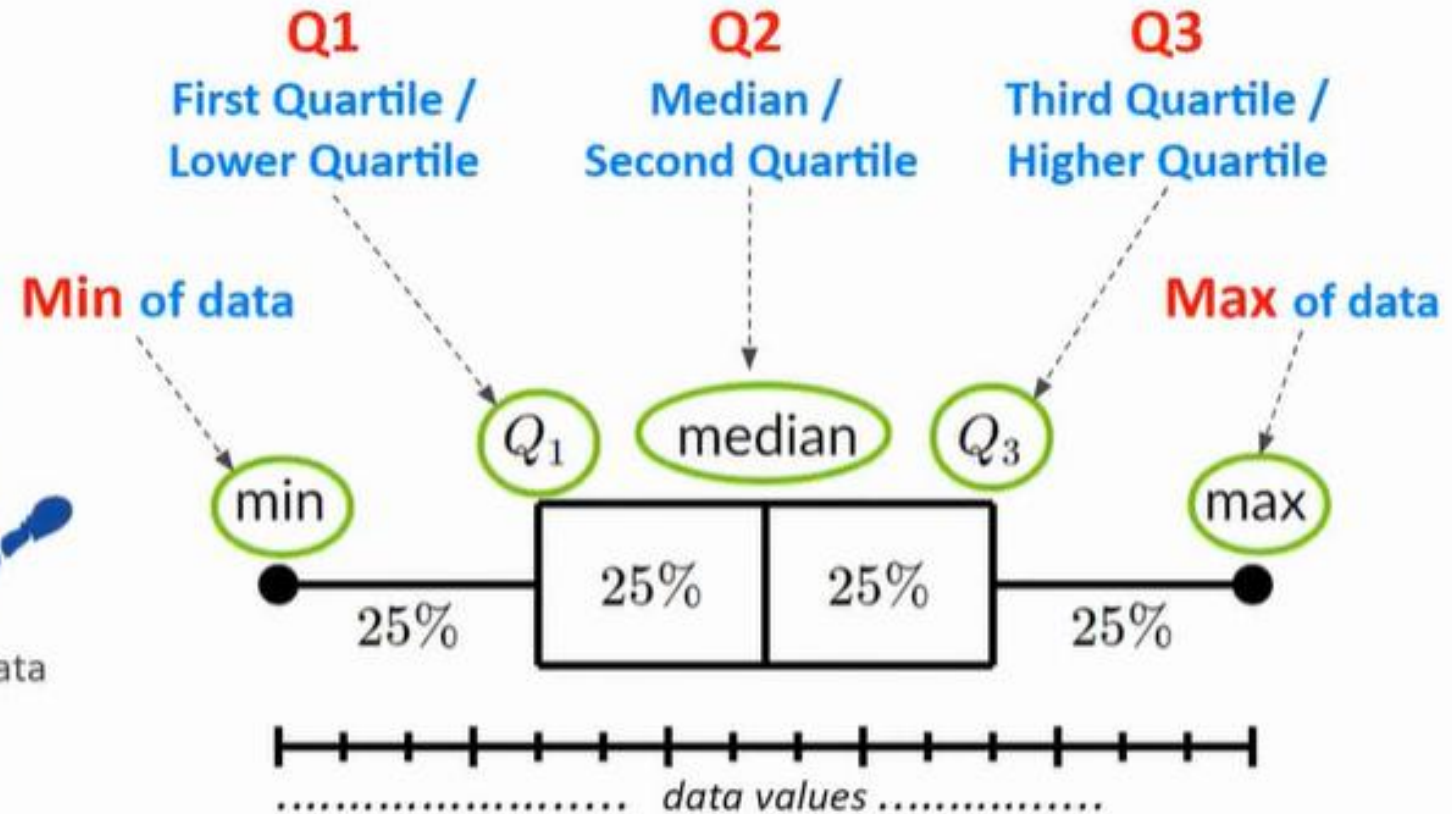
Five-Number Summary !



..... Used to display the distribution of data
as a **BOX / WHISKER PLOT**.

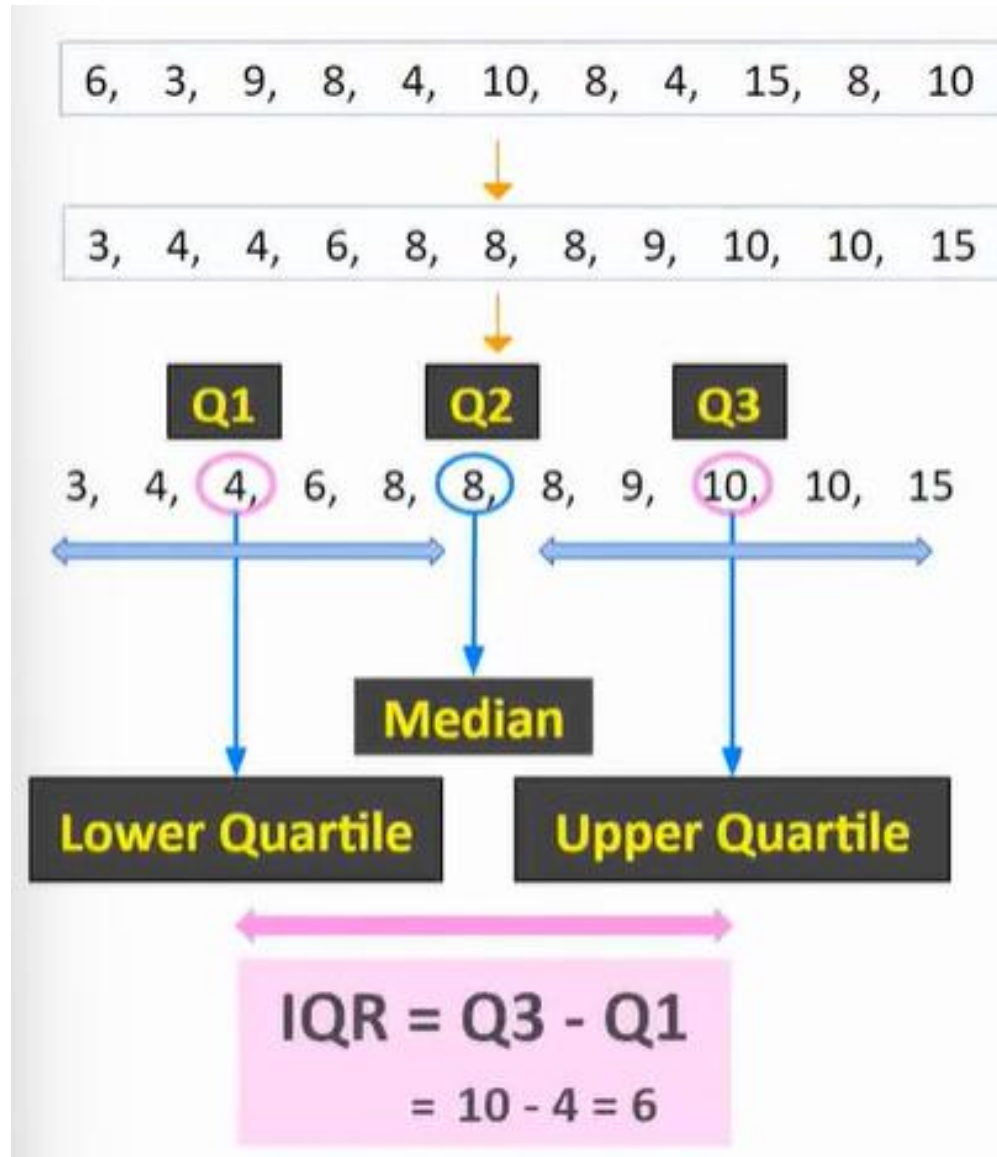
*(Divides the data into sections that each section
contain approximately 25% of the data)*

Mid-Spread (mid 50%) or, H-Spread.



$$\text{IQR} = Q_3 - Q_1$$

IQR = Upper Quartile - Lower Quartile
= 75th percentiles - 25th percentile



Steps to Compute IQR :

- 1.) For example, let's take a set of data
- 2.) Sort the data
- 3.) Find the Median (the number in the middle)
(If No. of values even, Median = Avg. of two mid-values)
- 4.) Again find the median in lower & upper half of the values.
 - Median in Lower half = Q1
 - Median in Upper half = Q3
- 5.) Interquartile Range, **IQR = (Q3 - Q1)**

$$\begin{aligned} \text{IQR} &= \text{Upper Quartile} - \text{Lower Quartile} \\ &= 75\text{th percentiles} - 25\text{th percentile} \end{aligned}$$

IQR is robust to Outliers !



Mid-Spread
(mid 50%) /
H-Spread



- If Data with Outlier
- If Skewed Data

Use IQR as a
Measure of
Dispersion



☐ Variance

☐ Std. Dev.



3.) Variance (σ^2)

- Avg. of Squared differences from the mean
- Not Good For Skewed Data
- Susceptible to Outliers.

N & N-1 Difference :
Bessel's Correction will
cover in another video.

Like **Mean**,
Variance computation too
take all the values in data.

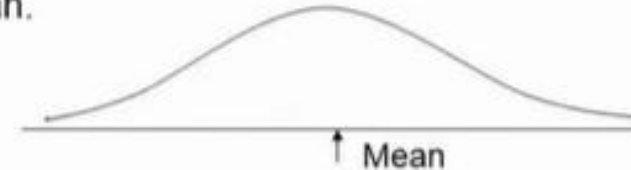
$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} \quad \text{Sample Variance}$$

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N} \quad \text{Population Variance}$$

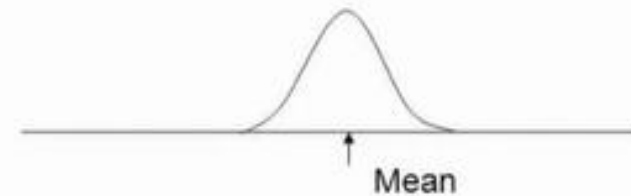
s^2 = sample variance
 x_i = value of i^{th} element
 $\bar{x}$ = sample mean
 n = sample size

σ^2 = population variance
 x_i = value of i^{th} element
 μ = population mean
 N = population size

The larger the variance, the further the individual cases are from the mean.

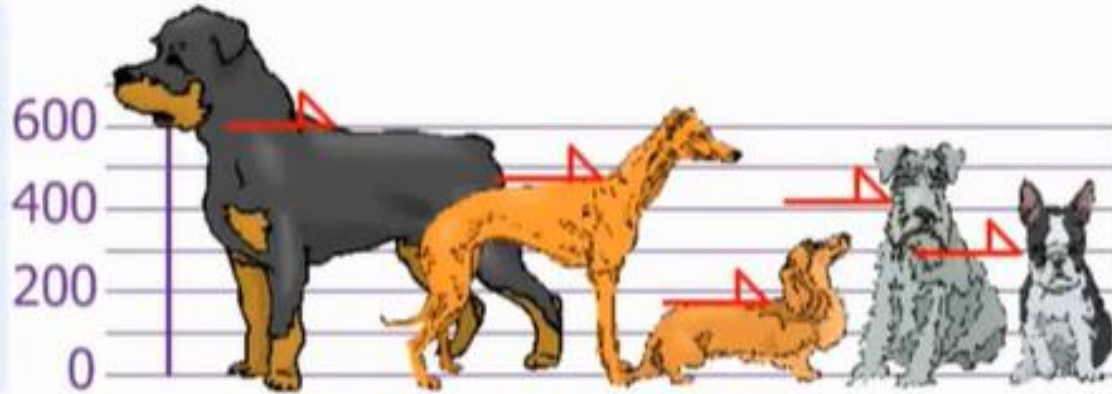


The smaller the variance, the closer the individual scores are to the mean.



VARIANCE : is the Average of the Squared Differences from the Mean.

Example : The heights (at the shoulders) are: 600mm, 470mm, 170mm, 430mm and 300mm.



Steps Computing
Variance :



1.) Find the Mean :

$$\begin{aligned}
 &= \frac{600 + 470 + 170 + 430 + 300}{5} \\
 &= \frac{1970}{5} \\
 &= 394
 \end{aligned}$$

The heights (at the shoulders) are: 600mm, 470mm, 170mm, 430mm and 300mm.



2.) Calculate diff. of each dog's height from the Mean.

(**Computed Mean = 394 mm**)

hence, 600-Mean, 470-meanSo on

Variance

$$\begin{aligned}
 \sigma^2 &= \frac{206^2 + 76^2 + (-224)^2 + 36^2 + (-94)^2}{5} \\
 &= \frac{42436 + 5776 + 50176 + 1296 + 8836}{5} \\
 &= \frac{108520}{5} \\
 &= 21704
 \end{aligned}$$

3.) Square each differences & then Average it.

So, Variance = 21704 **mm²**





Why Standard Deviation is
Preferred over Variance ?

In previous example, Unit of measurement of Original Data is "mm"

The heights (at the shoulders) are: 600mm, 470mm, 170mm, 430mm and 300mm.

And the Variance Obtained : 21704 mm²



Variance is no longer in the same unit as original data because of squaring the differences.

SOLUTION



Square root of Variance (Standard Deviation)

$$\sigma = \sqrt{\text{Variance}}$$

$$= \sqrt{21704} \text{ mm}^2$$

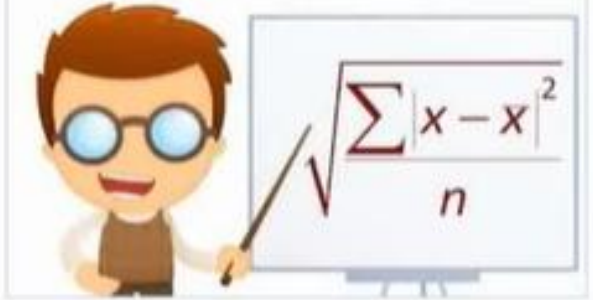
$$= 147.32 \text{ mm}$$

4.) Standard Deviation (σ)

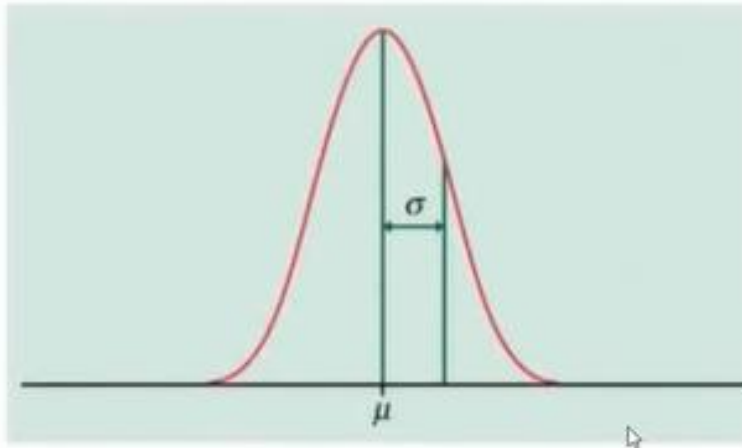
- Most Common , Stable & Widely used.
- Not Good, if Skewness in Data
- And Not Robust to Outliers.

Just Like, **Mean**
& **Variance** ...

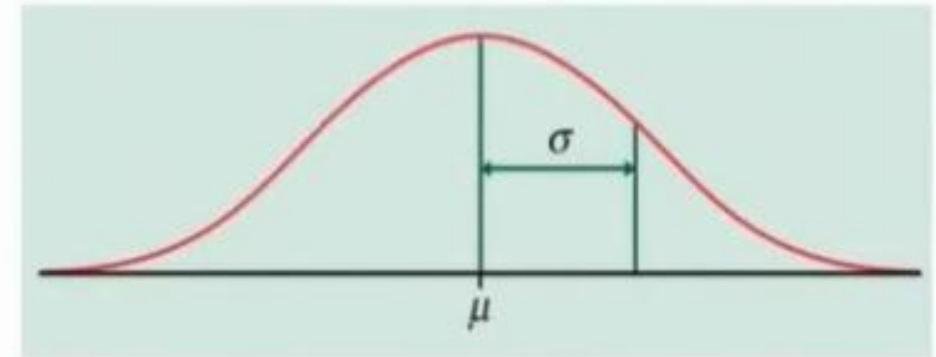
$$\sigma = \sqrt{\text{Variance}}$$



Low Std. Dev (σ) : Data are less spread out & are clustered around mean.



High Std. Dev (σ) : Data are more spread out.



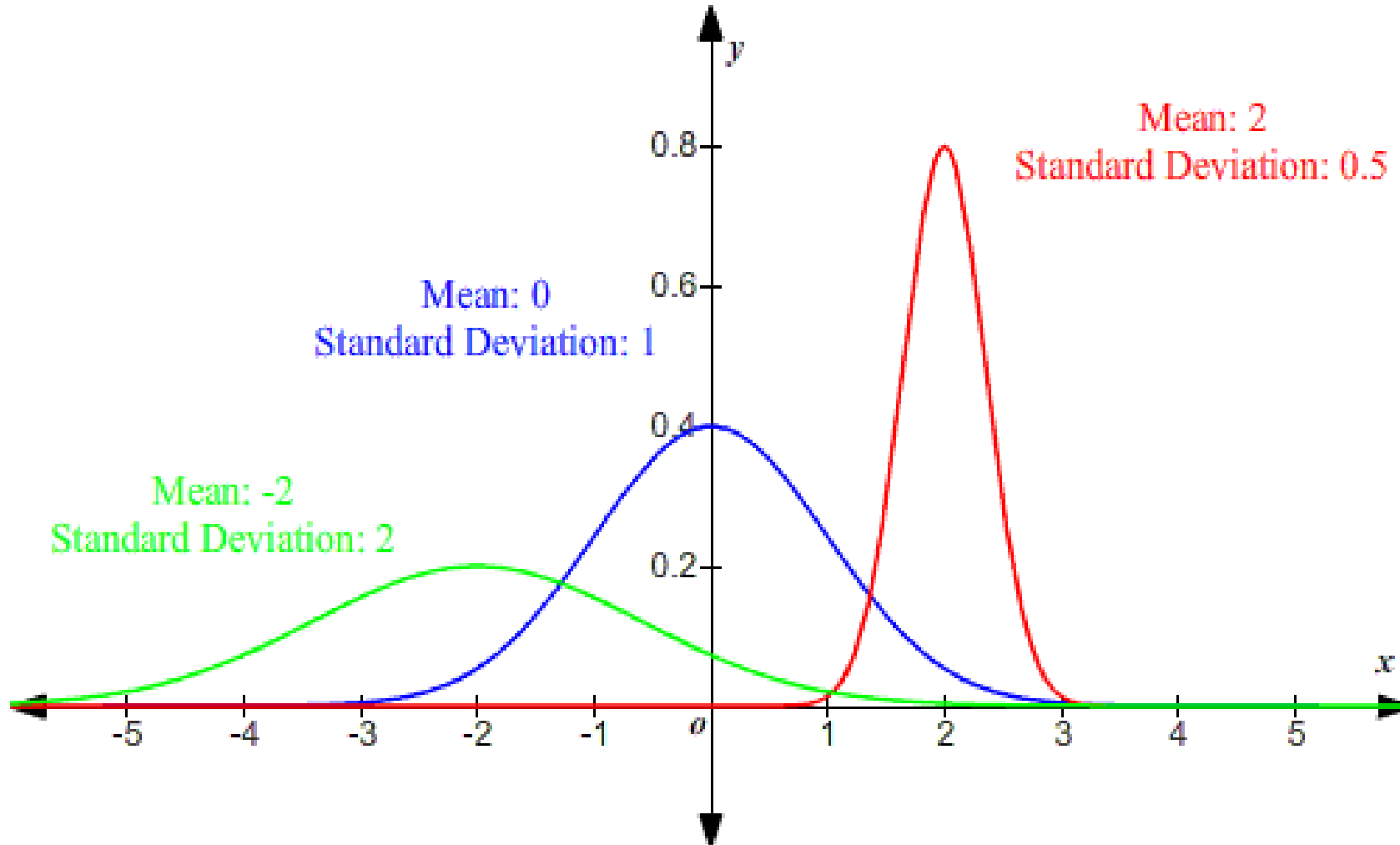
When to Use ?

- If Data with no or less Outlier
- If No or little Skewness in Data



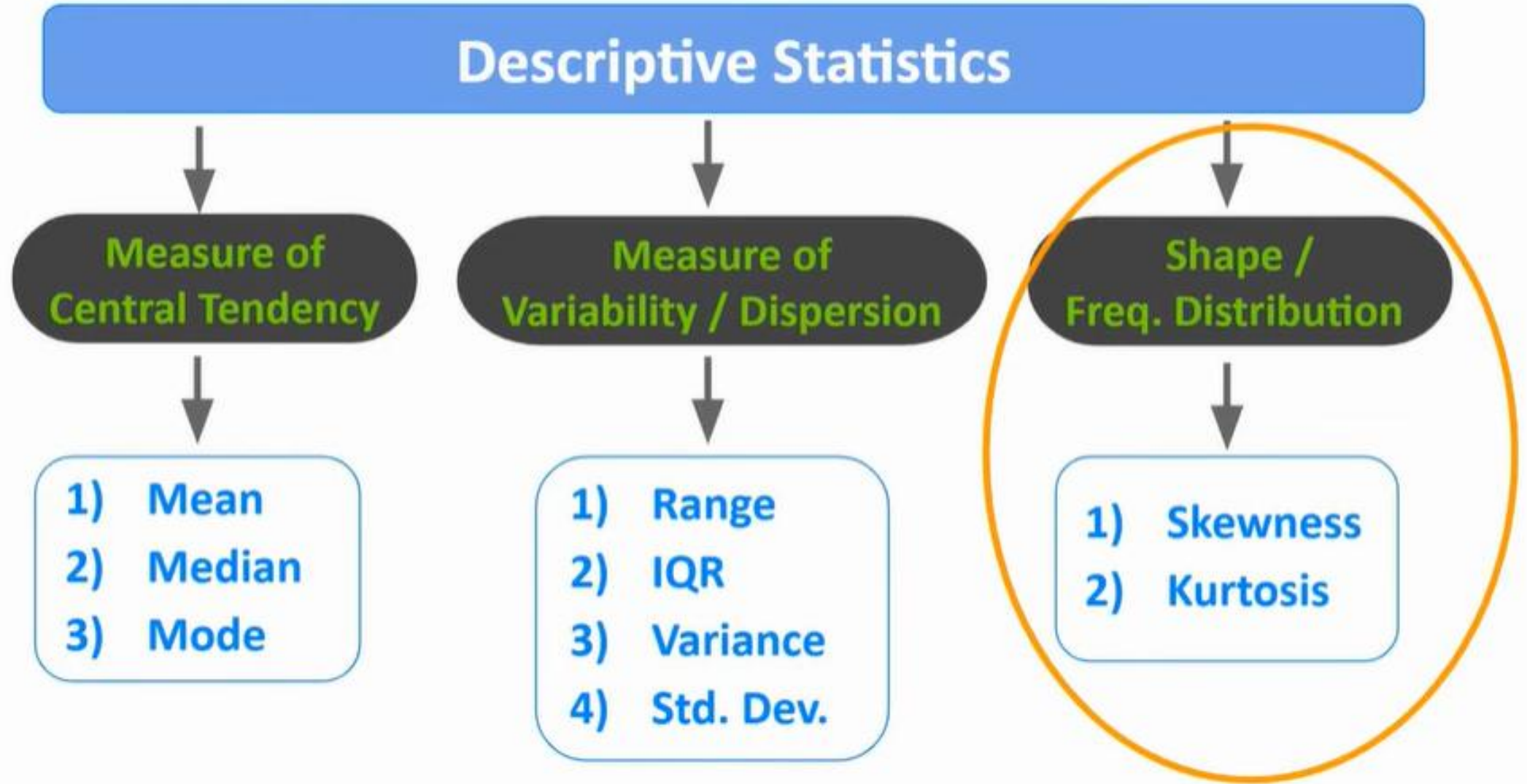
Use Std. Dev. ✓

What you can Infer?



Measures of Dispersion - Comparison

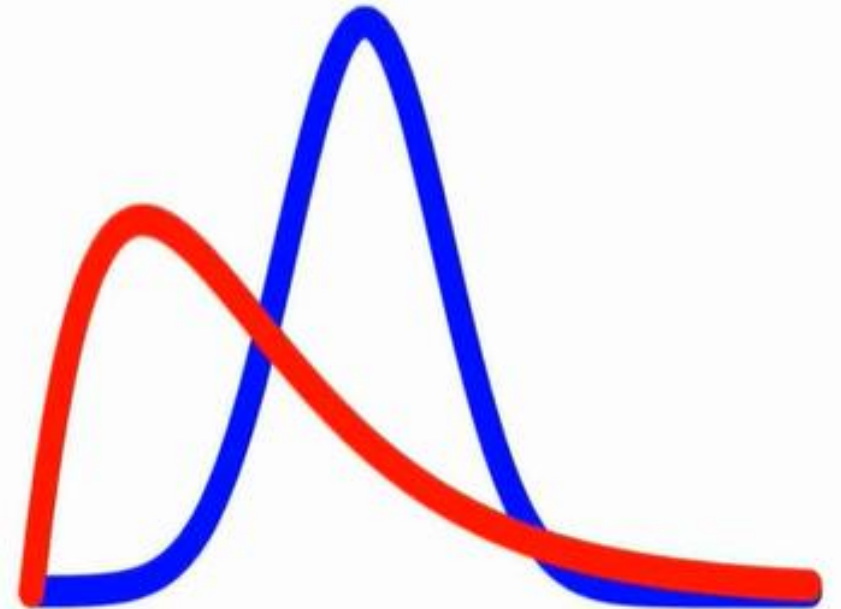
Different Measures of Dispersion	Formula	Data with Skewness	Data with Outliers	Pros	Remark
Range	MAX - MIN	✓	✓	Good for small size data	Doesn't take into account the distribution of data
IQR	Q3 - Q1	✓	✓	Good for Skewed data / data with Outliers	Considers only mid 50 % of total distribution of data
Variance (σ^2)	Avg. of Squared differences from the mean	✗	✗	-----	Variance doesn't end up having same unit as original data because of squaring the differences.
Std. Dev. (σ)	$\sigma = \sqrt{\text{Variance}}$	✗	✗	Most Stable & Widely used	-----



Distribution of Data

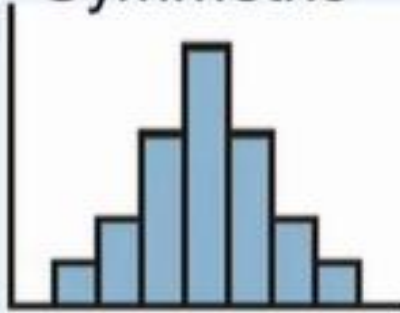


- What are all possible values for a variable in a dataset and how often they occur.
- These two statistics give insights into the shape of the distribution.

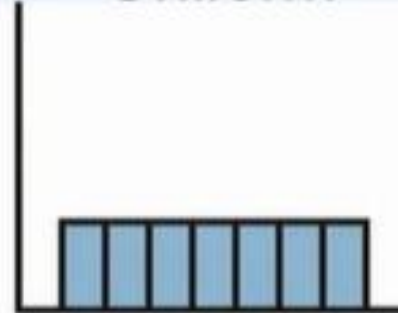


Diff. Shapes of Distribution :

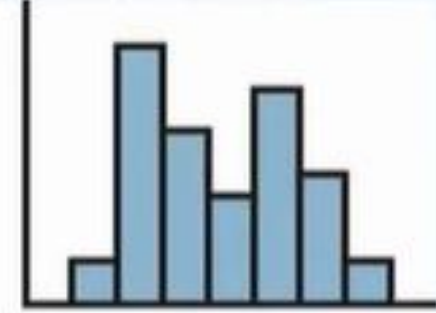
Symmetric



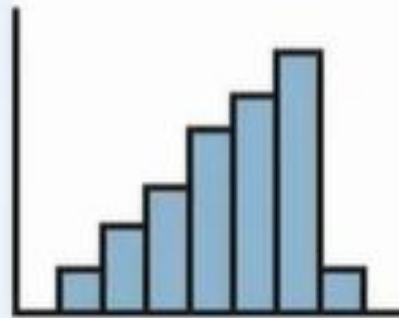
Uniform



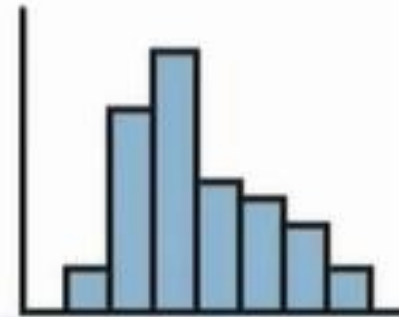
Bimodal



Skewed Left



Skewed Right



1.) Skewness

measures

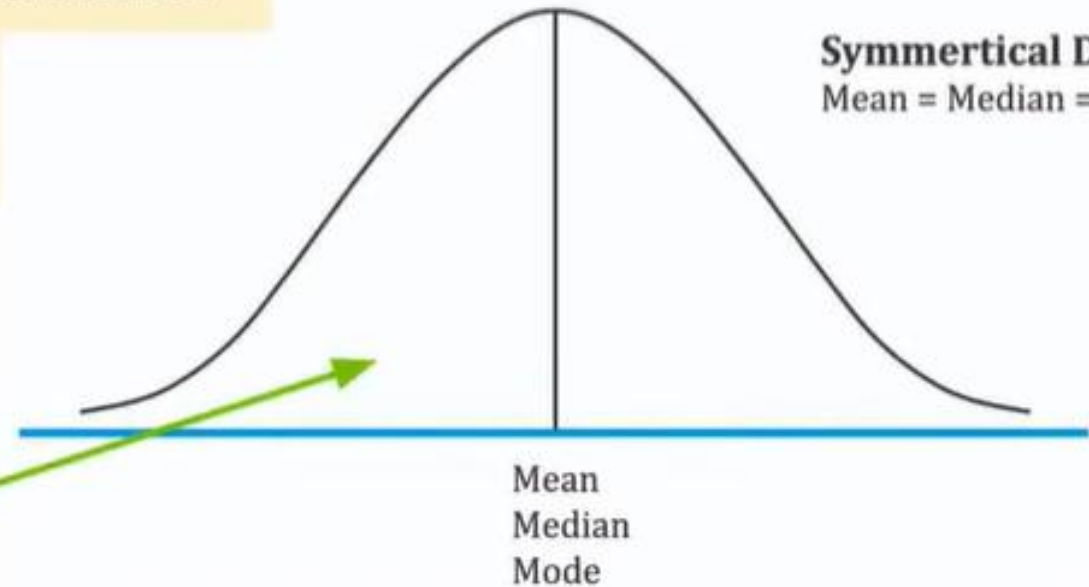


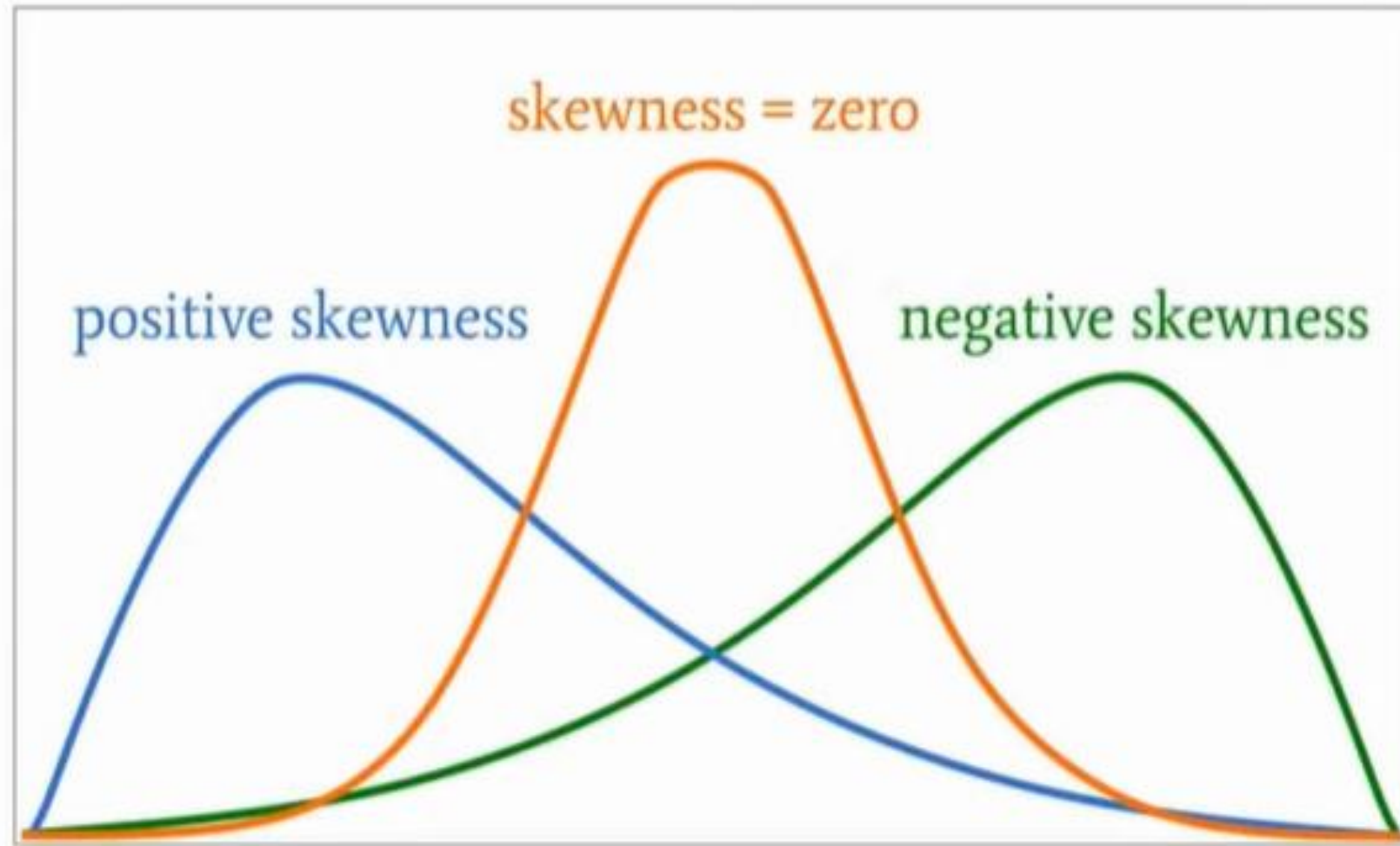
Lack of symmetry
in data distribution.

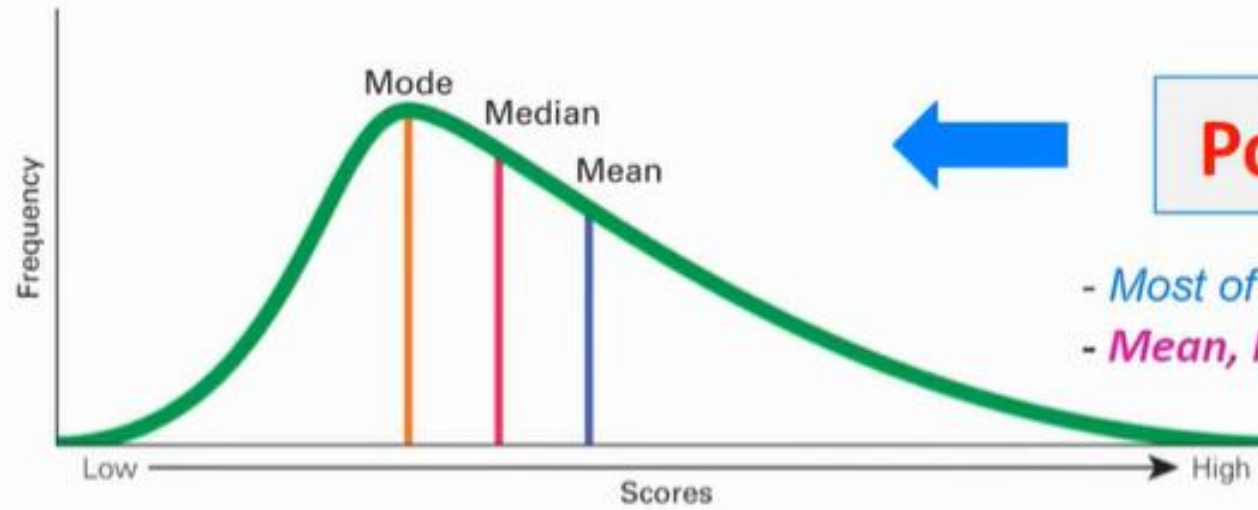
(Less Symmetric Data,
More the Skewness.)

- For Normal Distribution, Skewness is ZERO.
- Any kind of Symmetrical data shows Skewness close to ZERO.

Normal /
Symmetrical Distrib. :
Zero Skewness







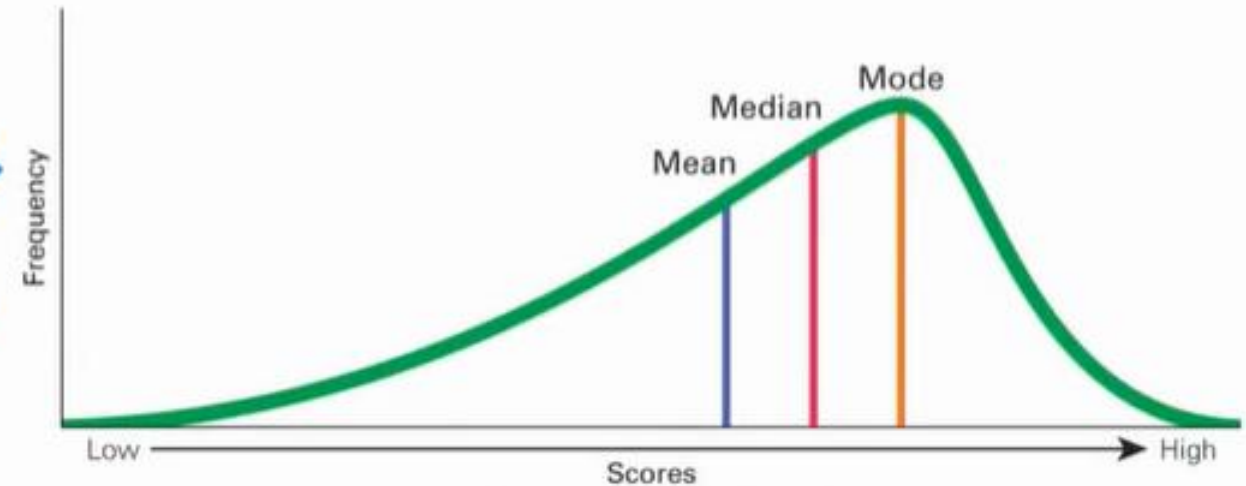
(a) Right-skewed distribution

Positive Skewness

- Most of the values clustered around left of the distrib.
- Mean, Median greater than the Mode.

Negative Skewness

- Most of the values are clustered around right.
- Mean, Median less than the Mode.



(b) Left-skewed distribution

What Value or Range of Skewness is Considered to be



Significant or Notable ?

General rule of thumb

If skewness is less than -1 or greater than 1 :



Highly skewed

If skewness is b/w -1 and -0.5 or b/w 0.5 and 1 :



Moderately skewed

If skewness is in b/w -0.5 and 0.5 :



Approximately symmetric

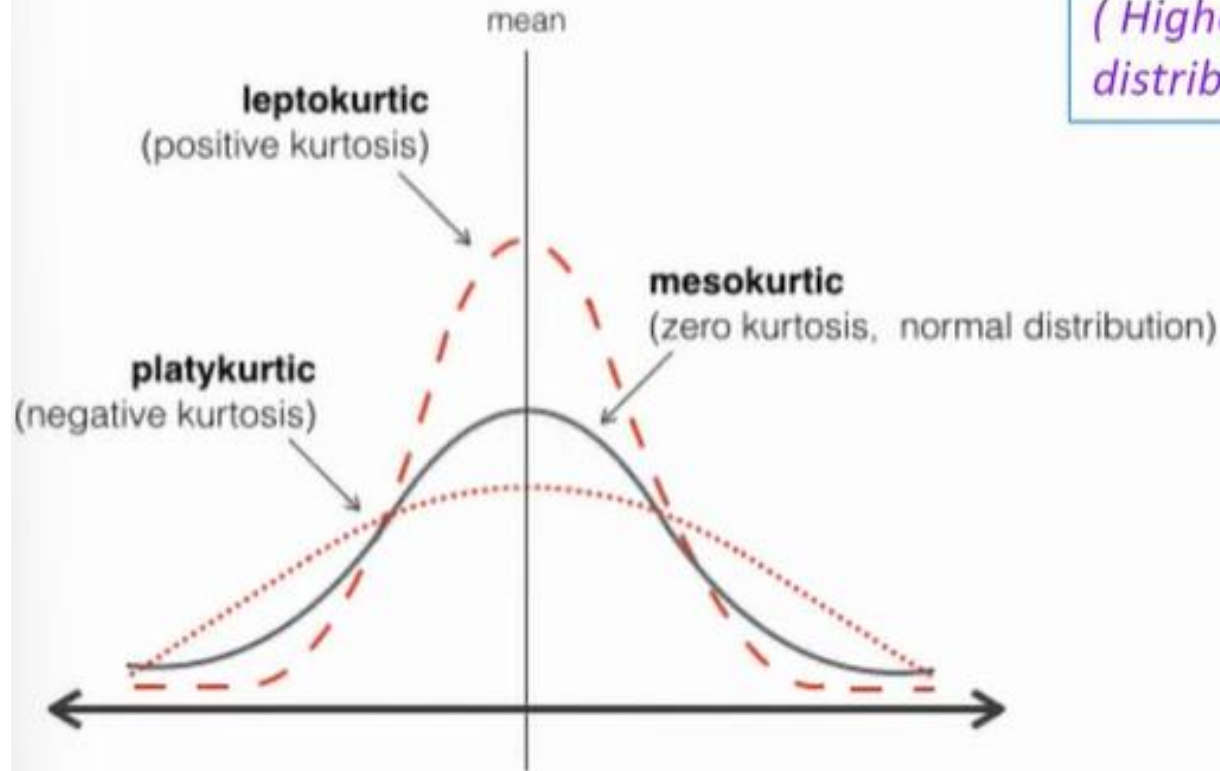
2.) Kurtosis :

measures



Whether data is heavy-tailed
or, light-tailed

(Higher the Kurtosis, Heavier the tail of data
distrib. or More the Outliers in data distribution.)

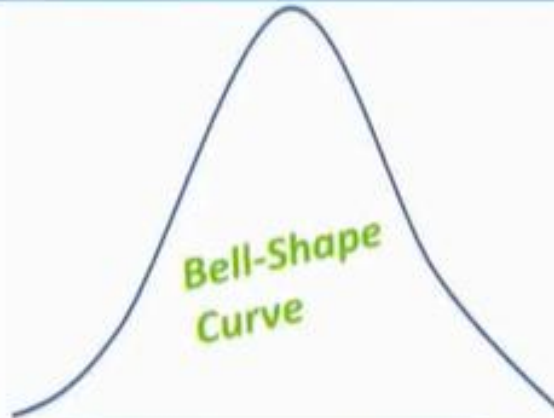


Kurtosis distribution :

- ★ Negative (-ve)
- ★ Zero
- ★ Positive (+ve)

Mesokurtic (Zero Kurtosis)

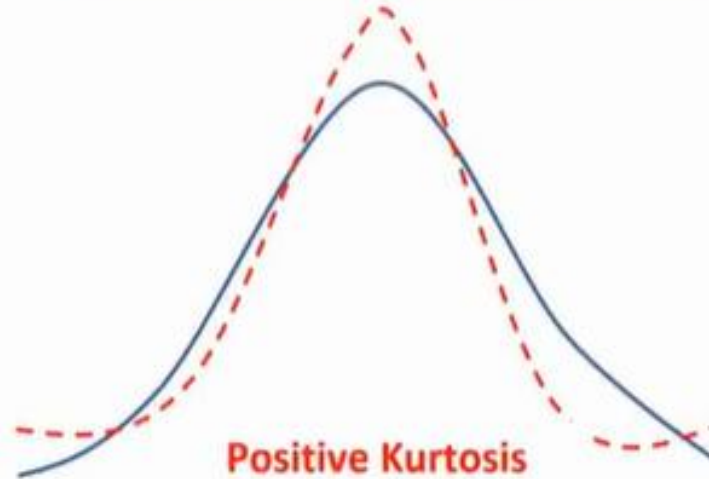
- Looks similar to the **Normal distribution Curve.**



Probability of having outliers or extreme values is zero or close to zero.

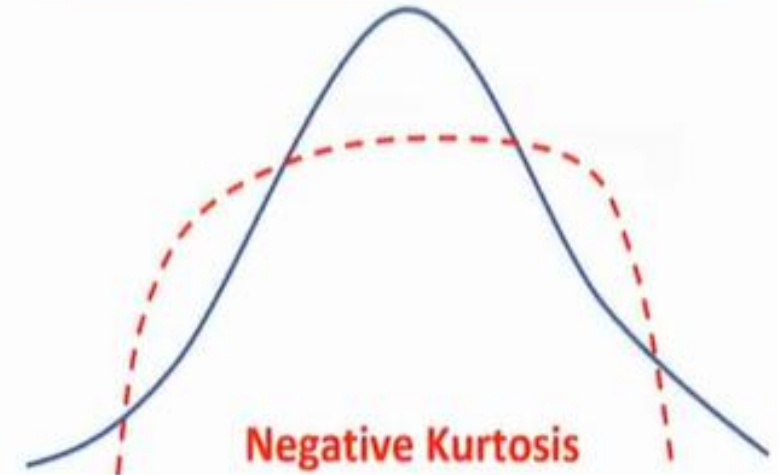
Leptokurtic (+Ve Kurtosis)

- Longer Distribution,
- Sharper peak & Heavier tails
- **More Outliers**



Platykurtic (-Ve Kurtosis)

- Shorter Distribution,
- Flatter peak & Thinner tails
- **Less Outliers**



**What Value of Kurtosis is
Considered to be**



**Significant or
Notable ?**

*..... already saw the
answer in previous
slide !*

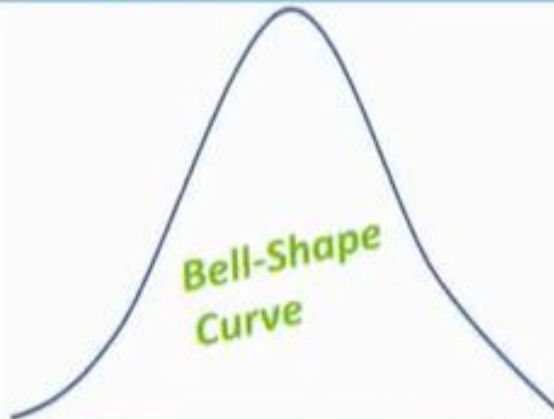
**High Kurtosis
More Outliers in Data**



Mesokurtic

(Zero Kurtosis)

- Looks similar to the **Normal distribution Curve.**



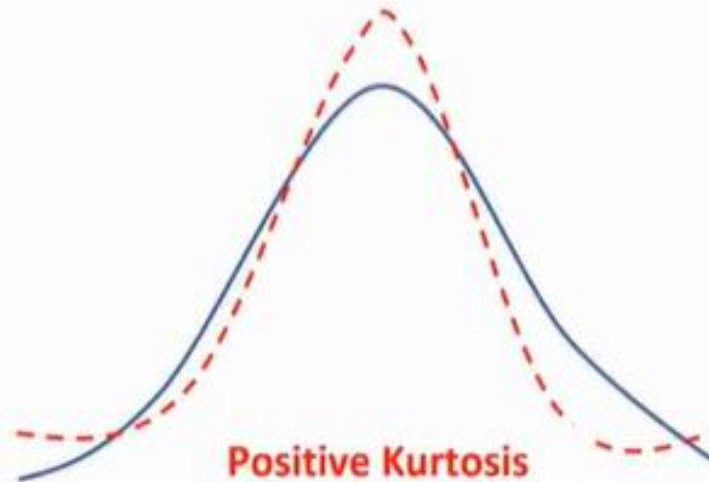
Probability of having outliers or extreme values is zero or close to zero.

Kurtosis = 0

Leptokurtic

(+Ve Kurtosis)

- Longer Distribution,
- Sharper peak & Heavier tails
- More Outliers**

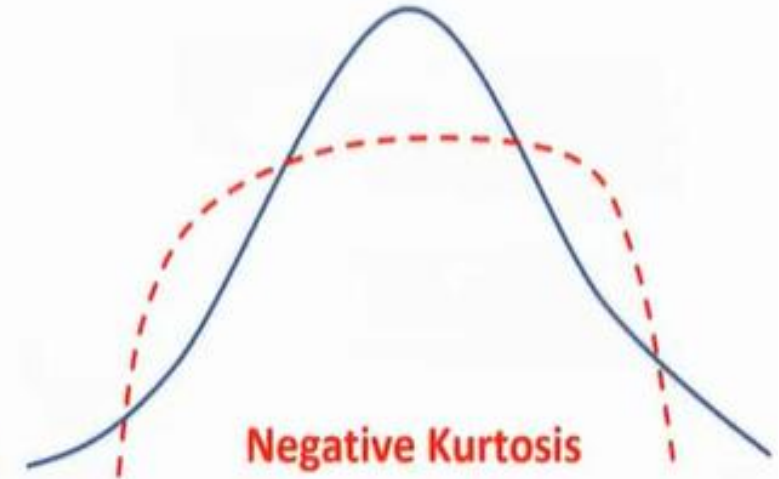


Kurtosis > 3

Platykurtic

(-Ve Kurtosis)

- Shorter Distribution,
- Flatter peak & Thinner tails
- Less Outliers**



Kurtosis < 3

When is Kurtosis Useful ?



- Only when its used in conjunction with Standard Deviation.

It is possible that a random variable might have a **high kurtosis** (bad), but the overall **standard deviation is low** (good).

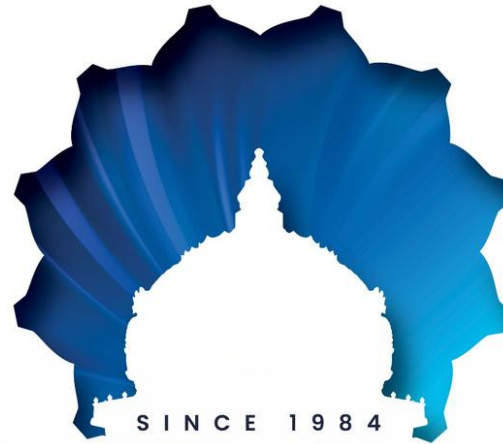
Conversely, one might see another random variable with a **low kurtosis** (good), but the overall **standard deviation is high** (bad).

References

<https://www.geeksforgeeks.org/data-science/statistics-for-data-science/>

<https://www.w3schools.com/statistics/>

<https://www.geeksforgeeks.org/maths/measures-of-dispersion/>



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THANK YOU
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